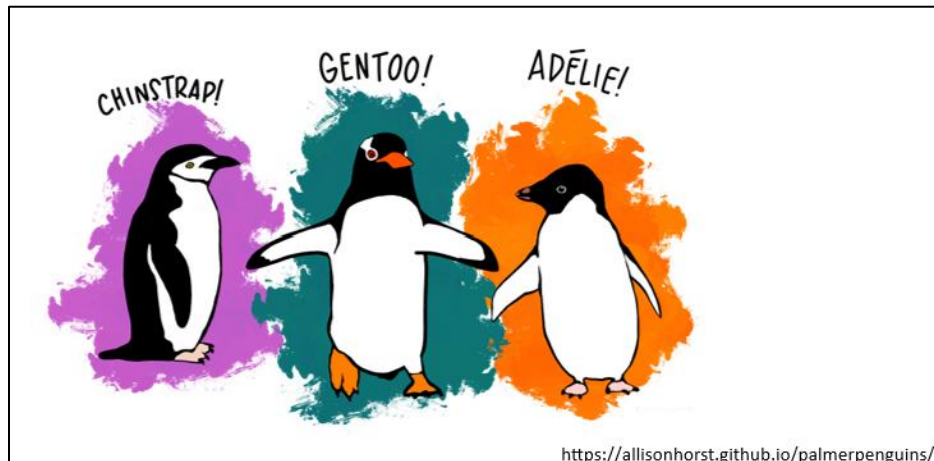
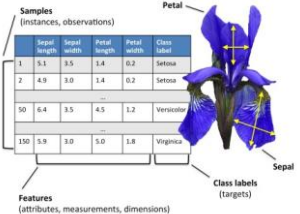


COP2034C Module 1 Programming Assignment

Welcome to your first Python programming assignment! In this course we will design Python programs that manipulate the [Palmer Penguins](#) dataset, based on research conducted from 2007–2009 by Dr. Kristen Gorman in collaboration with the Palmer Station Long Term Ecological Research (LTER) Program in Antarctica. The project tracked data on three penguin species observed in the Palmer Archipelago: Chinstrap, Gentoo, and Adélie.

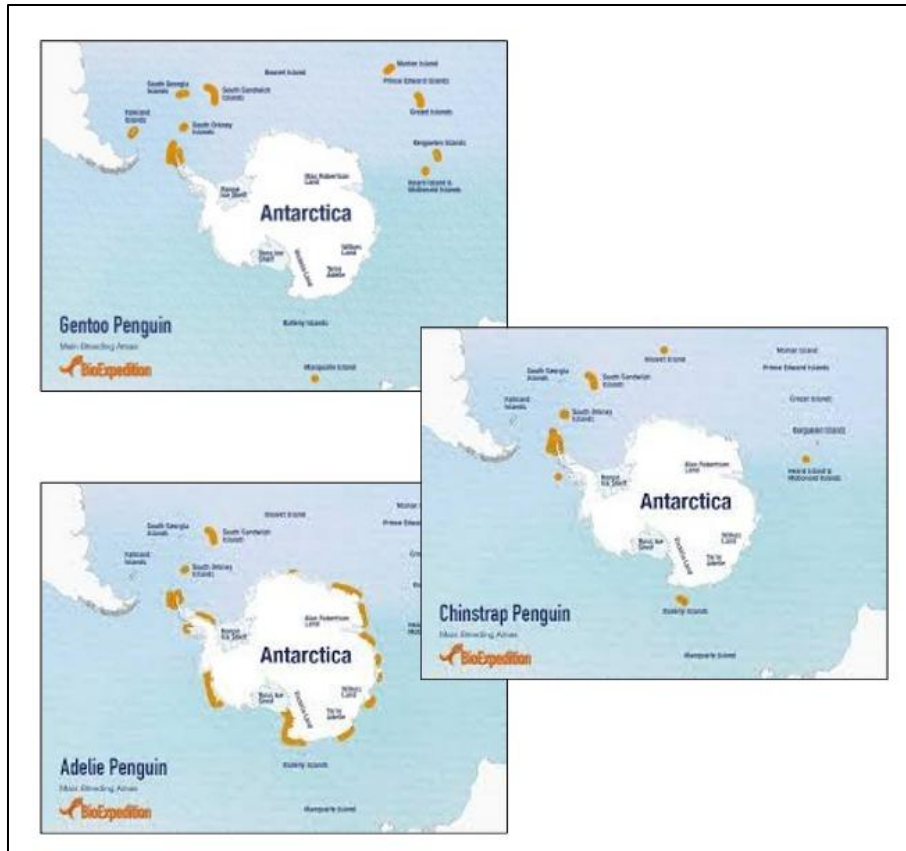


Historically there have been some well-known and well-used (overused) Data Science datasets such as those in the image shown here:

MNIST (1995)		IRIS: Fisher's Iris data set was introduced by the British statistician, eugenicist, and biologist Ronald Fisher in a 1936 paper.  <table><tr><th colspan="5">Samples (instances, observations)</th></tr><tr><th></th><th>Sepal length</th><th>Sepal width</th><th>Petal length</th><th>Petal width</th></tr><tr><td>1</td><td>5.1</td><td>3.5</td><td>1.4</td><td>0.2</td></tr><tr><td>2</td><td>4.9</td><td>3.0</td><td>1.4</td><td>0.2</td></tr><tr><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td></tr><tr><td>50</td><td>6.4</td><td>3.5</td><td>4.5</td><td>1.2</td></tr><tr><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td></tr><tr><td>150</td><td>5.9</td><td>3.0</td><td>5.0</td><td>1.8</td></tr></table>	Samples (instances, observations)						Sepal length	Sepal width	Petal length	Petal width	1	5.1	3.5	1.4	0.2	2	4.9	3.0	1.4	0.2	...	...	...	...	...	50	6.4	3.5	4.5	1.2	...	...	...	...	...	150	5.9	3.0	5.0	1.8
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Fashion MNIST (2017)																																										
MTCARS - 32 cars from the 1973-74 model year. Statistics about 1974 cars: fuel consumption, design, and performance.																																										

Enter Palmer Penguins!

- Data includes size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica.
- Freely available (“No Rights Reserved”).
- Provides a newer, fun dataset for data exploration & visualization.



Palmer Penguins Data Fields

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	StudyName	SampleNumber	Species	Region	Island	Stage	ID	ClutchCompletion	EggDate	CulmenLength-mm	CulmenDepth-mm	FlipperLength-mm	Bod Mass-g	Sex
2	PAL0708	1	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N1A1	Yes	11/11/2007	39.1	18.7	181	3750	MALE
3	PAL0708	2	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N1A2	Yes	11/11/2007	39.5	17.4	186	3800	FEMALE
4	PAL0708	3	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N2A1	Yes	11/16/2007	40.3	18	195	3250	FEMALE
5	PAL0708	4	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N2A2	Yes	11/16/2007					
6	PAL0708	5	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N3A1	Yes	11/16/2007	36.7	19.3	193	3450	FEMALE
7	PAL0708	6	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N3A2	Yes	11/16/2007	39.3	20.6	190	3650	MALE
8	PAL0708	7	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N4A1	No	11/15/2007	38.9	17.8	181	3625	FEMALE
9	PAL0708	8	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N4A2	No	11/15/2007	39.2	19.6	195	4675	MALE
10	PAL0708	9	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N5A1	Yes	11/9/2007	34.1	18.1	193	3475	

- **Species**
a factor denoting penguin species (Adélie, Chinstrap and Gentoo)
- **Island**
a factor denoting island in Palmer Archipelago, Antarctica (Biscoe, Dream or Torgersen)
- **CulmenLength_mm**
a number denoting bill length (millimeters)
- **CulmenDepth_mm**
a number denoting bill depth (millimeters)
- **FlipperLength_mm**
an integer denoting flipper length (millimeters)
- **BodyMass_g**
an integer denoting body mass (grams)

- **Sex**
a factor denoting penguin sex (female, male)
- **Year**
an integer denoting the study year (2007, 2008, or 2009)

Instructions

For this assignment, you will write a Python program that introduces the Palmer Penguin species. The program will use only basic output statements (the `print()` function). This assignment will show you how to create a simple program that will give you a quick success while helping you become familiar with the development environment we will use throughout the course.

Development Environment (all of the following steps will be demonstrated in class)

Log in to GitHub and fork the public repository provided by your instructor.

Then complete this assignment in [Google Colab](https://colab.research.google.com/) (<https://colab.research.google.com/>).

After your program executes correctly:

1. **Download** your notebook as a Python (.py) file.
2. **Upload** the .py file to your forked GitHub repository.
3. Take a screenshot showing the successful execution of your program in Google Colab and save it as a .png or .jpg file.
4. **Upload** the screenshot to your GitHub repository.

Do not submit the .ipynb notebook. The required submission for this assignment is the Python source file (.py) and the execution screenshot.

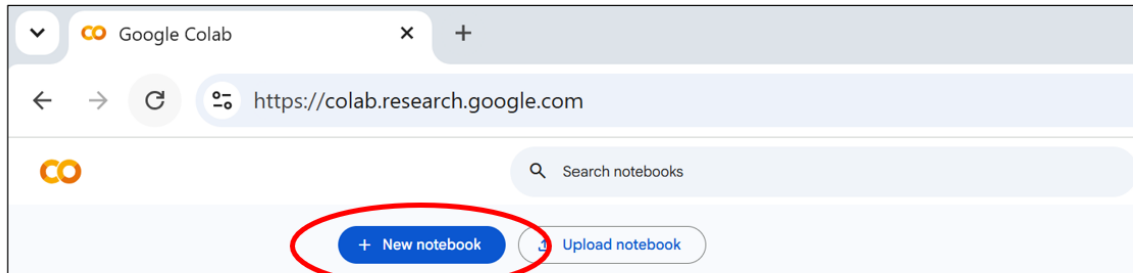
Expected Output

Your program must produce the following output exactly (including capitalization and punctuation).

```
Introducing the Palmer Penguins:
    Chinstrap!
    Gentoo!
and last but not least...
    Adelie!
There are a total of 3 penguin species in this dataset.
```

Walkthrough

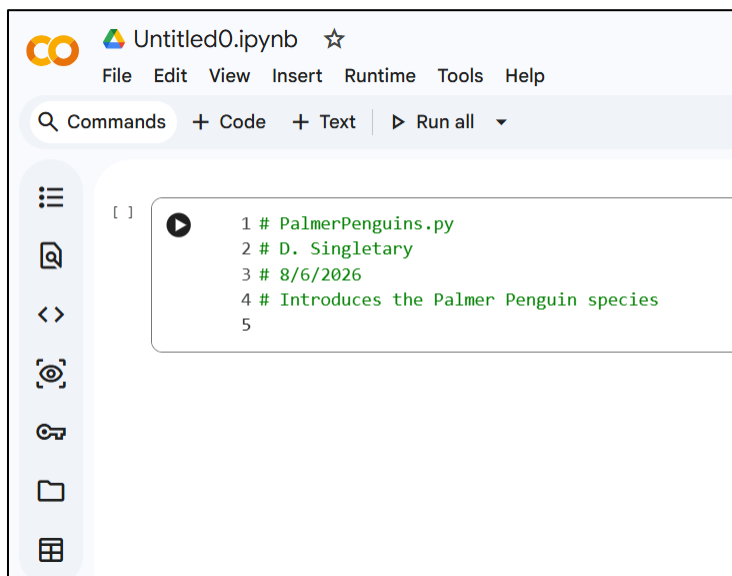
- Create a new notebook in Google Colab.



- Colab opens a new notebook with an empty code cell by default. Any code you enter in this cell can be executed at any time (and as many times as desired, with some restrictions that we will discuss later in the semester). Enter your Python code in this cell. The other type of cell is a text cell, which is used to add formatted notes, explanations, headings, or other documentation to your notebook.
- At the top of the file include the required four-line ID header.

```
# PalmerPenguins.py
# Your Name
# Submission Date
# Introduces the Palmer Penguin species
```

- Leave a blank line after the header for readability.



Constants

- In Python we commonly represent values that should not change for the life of the program using ALL UPPERCASE variable names. In the following code, we represent the three Palmer Penguin species and the total number of species using Python constants.

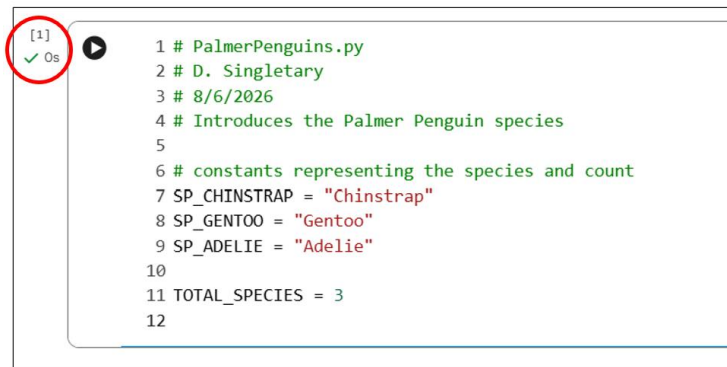
```
# constants representing the species and count
SP_CHINSTRAP = "Chinstrap"
SP_GENTOO = "Gentoo"
SP_ADELIE = "Adelie"

TOTAL_SPECIES = 3
```

- Using these **named constants** makes programs easier to read and maintain.

Incremental Development

- **Develop your program incrementally.** Write a small amount of code, run it, verify that it works, and then continue adding the next piece. Testing frequently makes it much easier to find and correct errors.
- Run the current code cell by clicking the **Run (▶)** button to the left of the cell or by pressing **Shift+Enter**. Although the program does not produce any output yet, running it confirms that your code executes without syntax errors before you continue. In the image below, the red circle highlights the green check mark, execution time, and execution count ([1]), indicating that the cell completed successfully without errors.



Output

- Use only the print() function for this assignment.
- Begin by displaying introductory text.

```
# output the species names with introductory text
print("Introducing the Palmer Penguins:")
```

- Display each penguin species exactly as shown in the expected output.
- You may use the tab escape sequence (`\t`) for indentation.

```
print("\t" + SP_CHINSTRAP + "!")
```

- Continue with the remaining output.

```
print("\t" + SP_GENTOO + "!")
print("and last but not least...")
print("\t" + SP_ADELIE + "!")
print()
print("There are a total of " +
      str(TOTAL_SPECIES) +
      " penguin species in this dataset.")
```

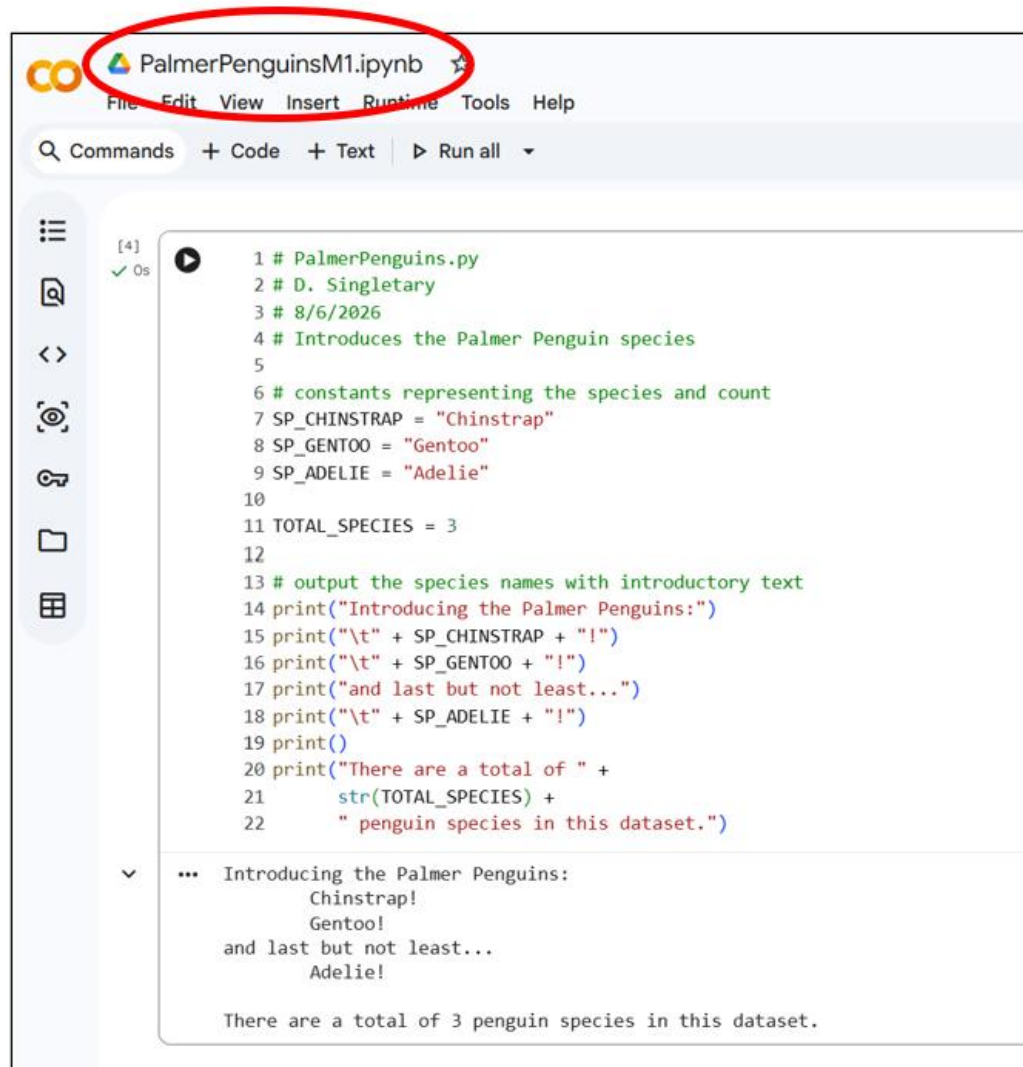
- Notice that `TOTAL_SPECIES` must be converted to a string using `str(TOTAL_SPECIES)` before concatenation.
- To print a blank line, we can just use `print()` with no arguments.

Readability

- In addition to whether your program runs correctly, all assignments in this course will also be graded on code readability. Keep your code readable by:
 - using consistent indentation.
 - separating logical sections with blank lines.
 - keeping lines reasonably short.
 - adding comments where appropriate.
 - choosing meaningful variable names.
 - using whitespace consistently around operators and after commas.

Saving and Downloading

- Give your notebook a name by clicking in the name field at the top and enter the new name:

A screenshot of a Google Colab notebook interface. The notebook title 'PalmerPenguinsM1.ipynb' is at the top, circled in red. Below the title is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A toolbar shows 'Commands', '+ Code', '+ Text', and 'Run all'. On the left is a sidebar with icons for file management. The main area contains a code cell with Python code for the Palmer Penguins dataset. The code includes comments, variable assignments for species names, and print statements. The output of the cell is visible below the code, showing the introductory text and the total number of species (3).

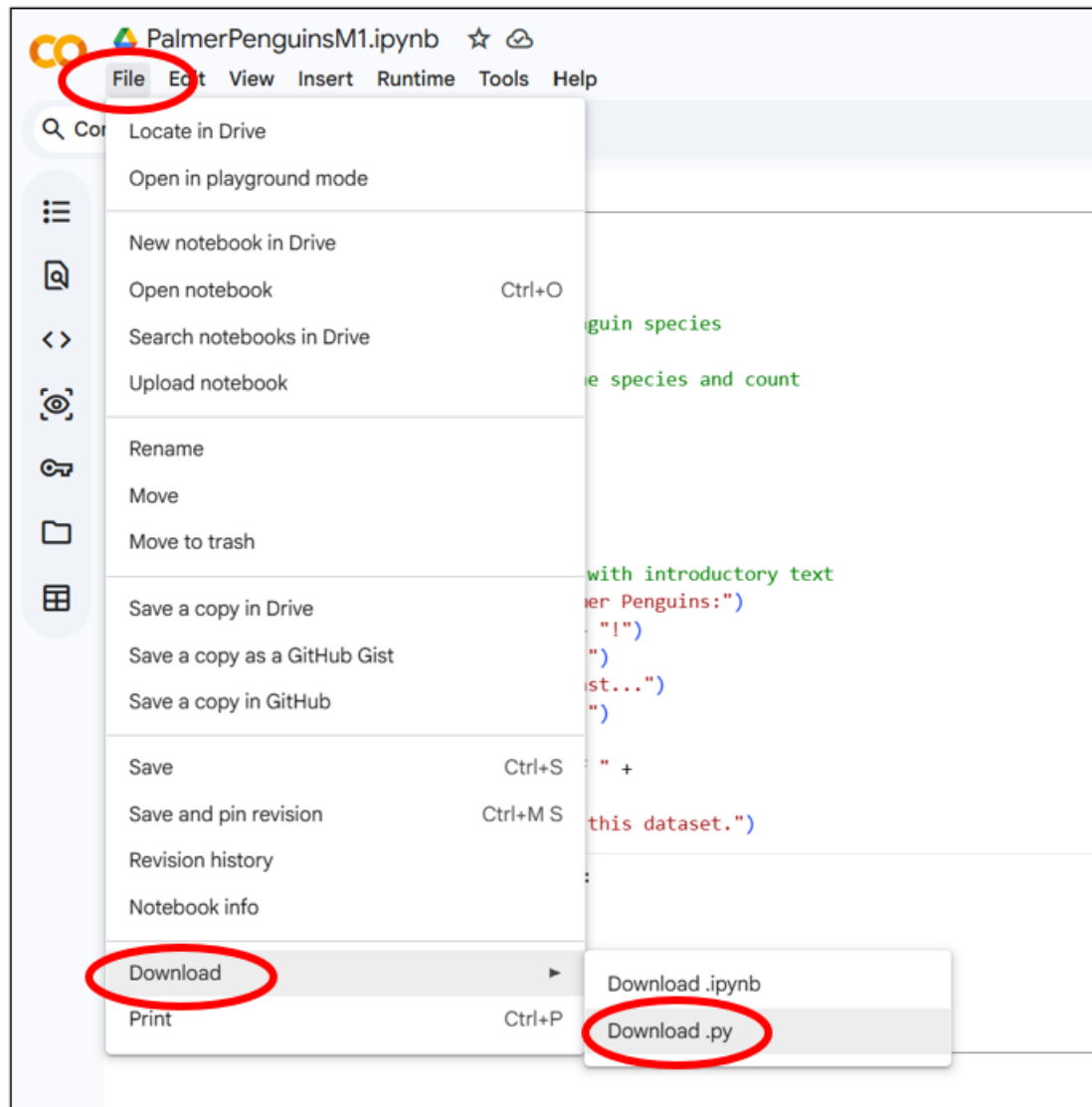
```
[4] ✓ 0s
1 # PalmerPenguins.py
2 # D. Singletary
3 # 8/6/2026
4 # Introduces the Palmer Penguin species
5
6 # constants representing the species and count
7 SP_CHINSTRAP = "Chinstrap"
8 SP_GENTOO = "Gentoo"
9 SP_ADELIE = "Adelie"
10
11 TOTAL_SPECIES = 3
12
13 # output the species names with introductory text
14 print("Introducing the Palmer Penguins:")
15 print("\t" + SP_CHINSTRAP + "!")
16 print("\t" + SP_GENTOO + "!")
17 print("and last but not least...")
18 print("\t" + SP_ADELIE + "!")
19 print()
20 print("There are a total of " +
21       str(TOTAL_SPECIES) +
22       " penguin species in this dataset.")

... Introducing the Palmer Penguins:
      Chinstrap!
      Gentoo!
and last but not least...
      Adelie!

There are a total of 3 penguin species in this dataset.
```

- Colab notebooks are auto-saved, but you can save them manually by using File/Save in the notebook's menu bar.

- We aren't as interested in saving the notebook as we are in saving the Python code. When your program is complete, download the Python program (do not download or submit the notebook). Download it using the File/Download/Download.py menu option as shown below. Check your downloads folder for the file, be sure it has a meaningful name. Since I called my notebook "PalmerPenguinsM1.ipynb", the downloaded file is called "palmerpenguinsm1.py".



Screenshot

- The final part of this assignment is to show what I call “proof of life”, i.e., a screenshot showing your output in the notebook for a program that has actually executed. We are only interested in the output, so here is the screenshot I took with the Windows snipping tool:

```
...   Introducing the Palmer Penguins:
      Chinstrap!
      Gentoo!
      and last but not least...
      Adelie!

      There are a total of 3 penguin species in this dataset.
```

- Don't include the source code in your image, I will be checking the .py file in your GitHub repo.
- I saved the file as “PalmerPenguinsM1.png”.

Submission

Upload the following two files to your forked GitHub repository:

- PalmerPenguinsM1.py
- Screenshot of the program executing successfully (.png or .jpg)

Submit the URL of your forked GitHub repository to the Canvas assignment.